

Ales Varabyou

Johns Hopkins University
3100 Wyman Park Drive, Room S216
Baltimore, MD 21211, USA
email: ales.varabyou@jhmi.edu
Website: <https://ccb.jhu.edu/people/ales.varabyou>
Languages: Belarusian, English, Polish, Russian

Education

2017 – 2023

Ph.D. (Genomics, CS)
Johns Hopkins University
Advisors: Dr. Steven Salzberg & Dr. Mihaela Perte
Thesis: Computational Study of Transcriptional landscapes from RNA-seq Data

2020

M.Sc.Eng. (Computer Science)
Johns Hopkins University

Advisors: Dr. Steven Salzberg & Dr. Mihaela Perte

2013-2017

B.A. *cum laude* (Biology & Computer Science)
Luther College
Honors Thesis: *Distributed Adaptive Object Recognition and Tracking*
Thesis advisor: Dr. Bradley Miller

Academic Positions

2026-present

Senior Scientist, Pacific Biosciences

2023-2026

Research Scientist at Perte Lab
Developing methods for analyzing transcriptome expression using RNA-seq data.
Building cell-type-specific genome annotations from large RNA-seq datasets.
Department of Biomedical Engineering, Johns Hopkins University

2023

Supervisor for an undergraduate project (CS). Topic: “Designing a Dynamic and Scalable Web Interface for Genome Annotations”

2023

Supervisor for an undergraduate project (BME). Topic: “Impact of Transcript Clustering on Downstream Expression Analysis”

2020

Teaching Assistant

2017

Research Assistant (Genomics)
Salzberg Lab, Institute of Genetic Medicine, **Johns Hopkins School of Medicine**

2015-2017

Research Assistant (Evolutionary Biology)
Luther College

2016

Teaching Assistant in Human Physiology
Luther College

Presentations And Talks

Varabyou A. (2023) Topics in Genomics. *Medical School Intensive. Guest Speaker*

Varabyou A. (2023) Precision Gene Catalogs: Segregating Noise and True Signals in RNA-seq Data for Comprehensive Genome Annotation. *Advanced Biomedical Computation (ABC), Seminar Speaker*

Varabyou A. (2022) Investigating Open Reading Frames in reference and novel transcripts using ORFanage. *Banbury Human Gene Annotation Meeting, Speaker*
Varabyou A., Pertea M., Salzberg S. (2019) Annotation-guided alignment of short RNA-seq reads for de-novo transcriptome assembly. *Genome Informatics*, Cold Spring Harbor, Poster

First Author Publications

Varabyou A., Artamonov M., Bolton D. (Upcoming: 2025). Full-length transcriptome analysis of clonal expansion and proviral integration during acute SIV infection at single-cell resolution. *bioRxiv*

Varabyou A., Artamonov M., Bolton D., Salzberg S., Pertea M. (2025). Comprehensive atlas of HIV-1 and SIV transcriptional diversity. *bioRxiv*, 2025-09

Varabyou A., Erdogdu B., Salzberg S., Pertea M. (2023). Investigating open reading frames in known and novel transcripts using ORFanage. *Nature Computational Science*, 10.1038/s43588-023-00496-1

Varabyou, A., Sommer, M. J., Erdogdu, B., Shinder, I., Minkin, I., ... & Pertea, M. (2023). CHESS 3: an improved, comprehensive catalog of human genes and transcripts based on large-scale expression data, phylogenetic analysis, and protein structure. *Genome biology*, 24(1), 249.

Varabyou A., Pockrandt C., Salzberg S., Pertea M. (2021). Rapid detection of inter-clade recombination in SARS-CoV-2 with Bolotie, *Genetics*, 10.1093/genetics/iyab074

Varabyou, A., Pertea, G., Pockrandt, C., Pertea, M. (2021). TieBrush: an efficient method for aggregating and summarizing mapped reads across large datasets. *Bioinformatics*, 10.1093/bioinformatics/btab342

Varabyou, A., Salzberg, S. L., & Pertea, M. (2021). Effects of transcriptional noise on estimates of gene and transcript expression in RNA sequencing experiments. *Genome Research*, 31(2), 301-308.

Liu, R., Yeh, Y. H. J., **Varabyou, A.**, Collora, J. A., Sherrill-Mix, S., Talbot, C. C., ... & Pollack, R. A. (2020). Single-cell transcriptional landscapes reveal HIV-1–driven aberrant host gene transcription as a potential therapeutic target. *Science Translational Medicine*, 12(543).

Contributing Author Publications

Olivera K., Wei Y., Varabyou A., Ho Y. C. (Upcoming May 2025). Spatial Mapping of SIV-infected Rhesus Macaque Lymph Node Environment During Rebound. *bioRxiv*

Erdogdu, B., **Varabyou, A.**, Hicks, S. C., Salzberg, S. L., & Pertea, M. (2024). Detecting differential transcript usage in complex diseases with SPIT. *Cell Reports Methods*, 4(3).

Amaral P., Carbonell-Sala S., De La Vega F. M., Faial T., Frankish A., Gingeras T., Guigo R., Harrow J. L., Hatzigeorgiou A. G., Johnson R., Murphy T. D., Pertea M., Pruitt K. D., Pujar S., Takahashi H., Ulitsky I., **Varabyou A.**, Wells C. A., Yandell M., Carninci P., Salzberg S. (2023). The status of the human gene catalogue. *Nature*, 622(7981), 41-47.

Sommer, M. J, Cha, S., **Varabyou, A.**, Rincon, N., Park, S., Minkin, I., Pertea, M., Steinegger, M., Salzberg, S. (2022). Structure-guided isoform identification for the human transcriptome. *Elife* 11, e82556.

Shifera, A. S, Pockrandt, C., Rincon, N., Ge, Y., Lu, J., **Varabyou, A.**, Jedlicka, A. E., Sun, K., Scott, A. L., Eberhart, C., Thorne, J. E., Salzberg, S. L. (2021). Identification of microbial agents in tissue specimens of ocular and periocular sarcoidosis using a metagenomics approach. *F1000Research*, 10(820).

Pertea, M., Shumate, A., Pertea, G., **Varabyou, A.**, Breitwieser, F.P., Chang, Y.C., Madugundu, A.K., Pandey, A. and Salzberg, S.L., (2018). CHES: a new human gene catalog curated from thousands of large-scale RNA sequencing experiments reveals extensive transcriptional noise. *Genome biology*, 19(1), pp.1-14.

Leadership and Volunteering

- 2022-2023** JHU Transportation advisory team. Designed and implemented a real-time tracking system for shuttle service delay reporting and accountability.
- 2022-2023** GRO Chair for Health and Wellness. Worked with other chairs and student organizations across campuses to create community events and help raise awareness of health and wellness resources on campus. Assisted individual students.
- 2021-2023** Volunteer as a panelist for academic misconduct hearings. Served on multiple academic hearings alongside faculty and members of school administration. Helped refine school policies and make decisions regarding cases of academic misconduct.
- 2022** Provost advisory team on healthcare. Served as a student representative of Homewood schools. Helped voice concerns regarding health insurance and advise on upcoming changes.
- 2022** Translated cybersecurity texts for Ukraine and Belarus during the onset of Russian invasion.
- 2020-2021** Aided refugees from Belarus. Held bi-weekly online sessions teaching spoken English and relevant computer science concepts. Provided guidance on immigration.
- 2020** Volunteer for the Emergency COVID-19 JHU hospital supplies team. Coordinated supply delivery and allocation between hospitals. Tracked employees and their shifts. Loaded and delivered supplies within Baltimore City.
- 2012-2018** Volunteer reviewer and selection committee member for the Belarus National UWC Committee.
- 2010-2014** Teaching assistant for Belarus Lyceum of Humanities. Supervised high school students on study abroad programs. Assisted teaching Computer Science, English, and Sciences.

Other Positions

- 2024-2025** Served as an invited external committee member for the MHRP hiring of PhD-level computational biology scientist
- 2018-2020** Volunteered consulting services to develop variant calling protocol at Aevus diagnostics.
- 2017-2018** Contributions and management of the MUMmer 4 suite on GitHub (~2,000 citations)

Reviewer for ISMB/ECCB 2021-2024, Nature, Molecular Biology and
Evolution, NAR, OUP Bioinformatics, Genome Biology, Computational and
Structural Biotechnology Journal, Springer Nature BMC
2015 - 2017 Workstation Administrator Assistant
Luther College
Supervisor: Matthew Hammen
2014 - 2016 Founder & President of the Luther College Robotics Research Club
Luther College
Academic Advisors: Dr. David Ranum, Dr. Bradley Miller & Dr. Kent Lee